

"New Continent Cup" 2026 Jiangsu Provincial University Student Computer Design
Competition Guide Vision Bridge Intelligence Person in Charge: Chen Junji Track:
Artificial Intelligence Practice

China Disabled Persons' Federation, Tencent Research Institute, National Rehabilitation Aids Research Center "Statistical Bulletin on the Development of the Disabled Persons' Cause" "Basic Situation Report on Internet Users with Visual Impairments in China" "Research on Aids Intelligence: Current Status of Visual Impairment Aids in China" Travel Conditions for the Visually Impaired There are nearly 17 million visually impaired individuals in China (including both fully blind and those with low vision). Strong desire for independent travel: $\geq 80\%$ Electronic guide devices are expensive and have poor adaptability, lacking large-scale usage. Those who can truly travel independently: only about 24%

Policy continues to release favorable signals, providing policy support and development direction for the human-machine collaborative blind navigation system project! Ministry of Industry and Information Technology: The "Law of the People's Republic of China on the Construction of a Barrier-Free Environment" stipulates that public service venues shall be equipped with necessary barrier-free facilities and assistive devices, with signs guiding barrier-free amenities, to provide barrier-free services for persons with disabilities and the elderly. China Disabled Persons' Federation: The "Notice of the State Council on Issuing the 14th Five-Year Plan for the Protection and Development of Persons with Disabilities" supports the development and application of new technologies in the field of information accessibility. It encourages the practical application of emerging technologies in areas such as blind navigation and voice control, and includes the integration of technology for disability assistance into the Action Plan for a Science and Technology Power, promoting the demonstration and application of science and technology in the fields of life health, artificial intelligence, and others in services for persons with disabilities.

易出错 识别不准 时效性缺失 无人工兜底 响应滞后 111 •
对低矮、动态及不规则障碍物识别精度低，易出现漏检、误判 •
现有地图仅支持静态路线，无法实时识别动态路障 •
纯AI语音导航易出现理解偏差与路线错误，无可靠纠错机制 •
路况信息无统一汇聚共享，视障者出行安全风险高 •
障碍识别与预警响应迟缓，无法实现实时避险 •
个性化路线无法语音记录，语义地址难以保存与复用
急需一款基于云边感知与时空数据融合的视障出行智能导盲系统

18426 Total number of dataset images Core categories include pedestrians, non-motor vehicles, steps, blind paths, potholes, roadblocks, traffic light status, etc.

ByteTrack多目标跟踪 (MOT, Multi-Object Tracking)
结合卡尔曼滤波与两阶关联策略，有效缓解了人流/车流
遮挡场景下的轨迹断裂。实测表明，引入ByteTrack后，
目标ID切换错误率 (IDSW) 下降超50%，轨迹断裂率降低约 45%。

BiFPN (Bidirectional Feature Pyramid Network, 双向特征金字塔网络) 是在EfficientDet (CVPR 2020) 中提出的多尺度特征融合架构, 其核心是通过双向信息流、加权特征融合以及优化跨层连接, 高效融合高低层特征, 显著提升了小目标与复杂场景的检测精度。

NCNN/OpenVINO INT8 PTQ Quantization essentially involves using a small amount of calibration data to determine the range and calculate the scaling factor (scale). Actual tests show that after INT8 quantization, the model size is drastically compressed to just 3.4MB. On the Raspberry Pi 5, the single-frame inference latency drops from over 200ms to 40ms (a speed increase of 4 to 5 times), perfectly meeting the real-time obstacle avoidance requirements at the edge.

Occlusion-Aware CutMix Introducing high-dimensional data augmentation strategies (Mosaic + CutMix). Real-world test results show that not only does the model's mAP improve by approximately 3.5% in absolute terms, but more critically, it significantly reduces background false positives (false alarm rate) by over 60%, greatly enhancing the system's "disturbance immunity" in complex street scenes.

Model Pruning removes redundant network weights, streamlining the structure while maintaining performance. This project employs channel-level pruning based on L1 norm, successfully reducing approximately 30% of parameters, significantly decreasing floating-point operations (FLOPs) and computational overhead. Knowledge Distillation utilizes a complex, high-precision model (teacher) to guide a lightweight model (student) in learning. This project adopts a Teacher model achieving an mAP of 84%, enabling the extremely compact Student model to inherit generalization ability. The overall mAP jumped from 67.4% to 81.7%.

Dual-model training is thorough, with stable convergence and no risk of overfitting. High detection accuracy has been achieved on both the training and validation sets. Obstacle detection accuracy is high.

Traditional Issues General large models lack real-time spatiotemporal awareness, making it difficult to directly understand the current location, speed, direction, and surrounding environment of visually impaired individuals, often leading to responses detached from the actual scenario. Dynamic Route Echoing Through multiplexing and segmentation, keywords such as "take me to" or "navigate to" are identified. Instead of entering the general casual conversation chain, the system directly switches to the navigation state machine, significantly reducing the time for the first word response and converting long text from the large model into short, broadcastable commands in real time. Before the speech is fed into the large model, the system injects the user's GPS coordinates, movement speed, direction, and front-view visual recognition results in real time, upgrading the AI from "general conversation" to a "spatiotemporal-aware assistant for obstacle avoidance navigation scenarios."

Multimodal AI Pre-generated Navigation Prompts for the Visually Impaired

Community submission: Manage road condition posts uploaded by volunteers via mobile snapshots. You can decide whether to adopt them as system data here.

Dispute Arbitration Tribunal: When multiple volunteers have disagreements over the annotation of the same obstacle, the system will automatically generate an arbitration work order.

The agent automatically parses the text and image road condition information reported by volunteers, extracts obstacle categories, locations, and risk characteristics, and generates structured data that can be stored and used for warnings.

AI route memory: View personalized semantic addresses created through voice by the blind client side, making it easier to locate and troubleshoot positioning drifts.

Cross-platform technology builds a volunteer mutual aid app, supporting volunteers to annotate and correct road condition information.

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